

SDG
14

Life Below Water

SDG 14 保育海洋生態

The ocean is closer than you think

海，比你以為的近

The Ocean Next Door

家門口的海

White Dolphins

白海豚

Plastic

塑膠

Our Actions

行動方案

27 km

to the Changhua coast

到彰化海邊的距離

< 50

Taiwanese white dolphins left

台灣白海豚剩不到

50 %

of your oxygen is from the sea

你呼吸的氧氣來自海洋

I grew up beside the Pacific Ocean.

我在太平洋旁邊長大。

In California, the sea was at the end of my street.

在加州，走到街底就是海。

I thought Taiwan would feel far from the ocean.

我以為台灣會離海很遠。

Then I looked at a map of Changhua.

後來我看了彰化的地圖。



Put up your hand — how many of you have seen the sea this year?

舉手看看——今年看過海的有多少人？

From California to Changhua

從加州到彰化

about 11,000 km west
向西飛約 11,000 公里

Huntington Beach

加州 · Dom 的家

Changhua, Taiwan

彰化 · 我們在這裡



Everything between my home and yours is ocean. It is one ocean, not two.

我家和你家之間全部都是海。那是同一片海，不是兩片。



Stand Up If...

如果是你，就站起來！

1

Stand up if you have eaten oysters.

吃過蚵仔的，站起來。

2

Stay standing if you have been to the Changhua coast.

去過彰化海邊的，繼續站著。

3

Stay standing if you have ever picked up litter on a beach.

曾經在海邊撿過垃圾的，繼續站著。

Look around. Those oysters grew 27 km from this classroom.

看看四周。你吃的那些蚵仔，就長在離這間教室 27 公里的地方。

27 km

from your school to the sea

從你的學校到海邊

Tianwei is inland — but Fangyuan is less than an hour away by car.

田尾是內陸鄉，但開車到芳苑不用一小時。

?

Does living inland mean the ocean is not your problem?

住在內陸，海洋就不關你的事嗎？



The River Runs Past Your School

那條溪，就從你們學校旁邊流過

The Dongluo River — the old Zhuoshui River — runs along the edge of Tianwei, through Pitou and Erlin, and reaches the Taiwan Strait at Fusing. A plastic wrapper dropped into a drain here can be in the sea in a few days.

東螺溪（舊濁水溪）沿著田尾鄉界流過，經過埤頭、二林，最後在福興鄉出海。在這裡掉進水溝的一個塑膠包裝，幾天後就可能在海裡。

river
河流

drain
水溝

estuary
出海口

downstream
下游

< 50

Taiwanese white dolphins are left in the world

全世界的台灣白海豚，剩下不到 50 隻

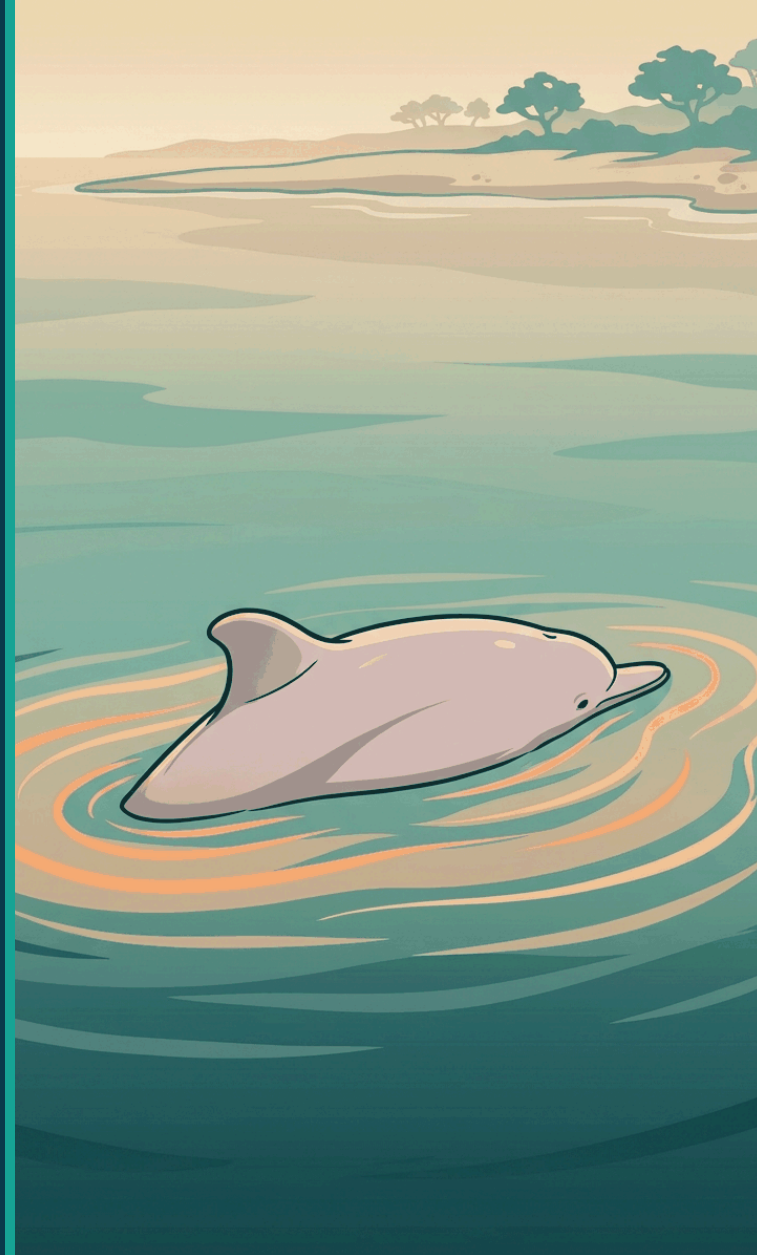
**Fishers call them Matsu fish. They live only along
Taiwan's west coast.**

漁民叫牠們「媽祖魚」，只住在台灣西海岸。

?

**This school has more students than the world has white
dolphins.**

這所學校的學生，比全世界的白海豚還多。



How much of the oxygen you are breathing right now came from the ocean?

你現在呼吸的氧氣，有多少來自海洋？

A About 10%

大約 10%

B About 25%

大約 25%

C About 50%

大約 50%

D Almost none

幾乎沒有

 Tap any option to reveal · 點任一選項公布答案

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你現在呼吸的氧氣，有多少來自海洋？

A About 10% 大約 10%

B About 25% 大約 25%

C About 50% 大約 50%

D Almost none 幾乎沒有



Half. Every second breath you take was made by the sea.

一半。你每呼吸兩次，就有一次是海洋給的。

8M

tons of plastic enter the ocean every year

每年有 800 萬噸塑膠進入海洋

About one garbage truck emptied into the sea every minute, all year.

大約等於每分鐘一台垃圾車倒進海裡，全年無休。

?

Ships, or the land — where does most of it come from?

船上還是陸地？大部分是從哪裡來的？



How long does a plastic bottle take to break down in the sea?

一個塑膠瓶在海裡要多久才會分解？

A About 5 years

大約 5 年

B About 50 years

大約 50 年

C About 450 years

大約 450 年

D It never fully
disappears

永遠不會完全消失

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About 450 years — and it never really vanishes. It just breaks into microplastic.

大約 450 年——而且它其實不會消失，只是碎成塑膠微粒。

True or False?

真的還假的？手比 ○ 或 ✕

Most ocean plastic comes from ships.

海洋裡的塑膠大多來自船隻。

Fangyuan has the last and largest natural tidal flat in Taiwan.

芳苑有全台灣最後也最大的一片原始潮間帶。

Microplastic has been found in the fish people eat.

人吃的魚身上已經驗出塑膠微粒。

One school is too small to change anything.

一所學校太小，改變不了什麼。



Tap each one to reveal · 點每一題公布答案

True or False?

真的還假的？手比  或 

Most ocean plastic comes from ships.

海洋裡的塑膠大多來自船隻。

 **FALSE**

Most of it comes from the land — through rivers and drains.

大部分來自陸地，經由河流和水溝流進海裡。

Fangyuan has the last and largest natural tidal flat in Taiwan.

芳苑有全台灣最後也最大的一片原始潮間帶。

 **TRUE**

It is in your county — and it is still working farmland for oysters.

就在你們縣內，而且到現在還是活生生的蚵田。

Microplastic has been found in the fish people eat.

人吃的魚身上已經驗出塑膠微粒。

 **TRUE**

What we throw into the sea comes back to our own plates.

我們丟進海裡的東西，會回到自己的餐盤上。

One school is too small to change anything.

一所學校太小，改變不了什麼。

 **FALSE**

Fangyuan's ox-cart oyster farming survives because local people refused to let it go.

芳苑的海牛採蚵能留到今天，就是因為在地人不肯放手。



Tap each one to reveal · 點每一題公布答案

12 Oxen Left

只剩下 12 頭牛

For sixty years, Fangyuan farmers have walked oxen onto the mudflat to harvest oysters. Machines took over. About twelve working oxen remain — the last place on Earth.

六十年來，芳苑蚵農趕著黃牛走進潮間帶採蚵。機械取代之後，只剩約十二頭海牛——地球上最後一個。

tidal flat
潮間帶

oyster
牡蠣・蚵仔

heritage
文化資產

disappear
消失



30-Second Challenge

30 秒挑戰

In 30 seconds, shout out as many sea creatures in English as you can. I will count!

30 秒內，用英文喊出越多海洋生物越好，我來數！

dolphin 海豚

oyster 蚵仔

crab 螃蟹

jellyfish 水母

seaweed 海藻

turtle 海龜

shrimp 蝦子

whale 鯨魚

Class record to beat: 18

要打破的班級紀錄：18 個



Which Helps the Ocean Most?

哪一件事對海洋幫助最大？

Rank these four from most helpful to least. Argue for your answer!

把這四件事從幫助最大排到最小。說出你的理由！

1

Refusing a plastic straw once

拒絕一支塑膠吸管

2

Picking up litter near your school drain

撿起學校水溝旁的垃圾

3

Getting your whole family to use one less plastic bag a day

讓全家每天少用一個塑膠袋

4

Telling ten people what you learned today

把今天學到的告訴十個人

Reveal 公布



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There is no single right answer — but the last two multiply. One person's habit becomes many.

沒有標準答案——但後面兩個會「放大」。一個人的習慣會變成很多人的習慣。

Turn and Tell Your Partner

轉過去，告訴你的同學

If the white dolphins disappear, what will Changhua have lost?

如果白海豚消失了，彰化會失去什麼？

If the dolphins disappear, Changhua will lose _____.

如果白海豚消失了，彰化會失去 _____。



One minute each. Then I will ask two of you.

每人一分鐘。然後我會請兩位分享。

Dom on Life Below Water

Dom 談海洋



Watch — then tell me the one thing you will protect.

看完之後，告訴我你決定要保護的那一件事。

3 Things You Can Do This Week

這週你可以做的 3 件事

1

Carry your own bottle and bag for one week.

帶自己的水壺和袋子，撐一個星期。

2

Pick up one piece of litter near a drain on your way home.

回家路上，在水溝旁撿起一件垃圾。

3

Ask your family to visit the Changhua coast — and see it for yourself.

跟家人提議去一趟彰化海邊，親眼看看。

Say It With Me

跟著我唸

Life below water.

水面下的生命。

The ocean gives me half my air.

海洋給了我一半的空氣。

The river carries what I drop.

我丟的東西，河會帶走。

Fewer than fifty are left.

剩下不到五十隻。

I can be the change.

我可以成為那個改變。



Game Time! 遊戲時間

BONUS 加碼

Q3

How much of the Earth's surface is ocean?

地球表面有多少比例是海洋？

A About 40%

大約 40%

B About 55%

大約 55%

C About 71%

大約 71%

D About 90%

大約 90%



Tap any option to reveal · 點任一選項公布答案

How much of the Earth's surface is ocean?

地球表面有多少比例是海洋？

A About 40% 大約 40%

B About 55% 大約 55%

C About 71% 大約 71%

D About 90% 大約 90%



About 71%. We call it Earth — but it is mostly ocean.

大約 71%。我們叫它「地球」，但它其實大部分是海。



Leave a Message for Dom

留言給 Dom

Go to our School Tour page and find the Nanzhen Elementary column. No sign-up, no login — tell me what you will protect.

到我們網站的 School Tour 頁面，找到南鎮國小那一欄，免註冊、免登入，告訴我你想保護什麼。

mycultureconnect.org/dom-school-tour.html

I will protect _____ because _____.

我想保護 _____，因為 _____。

**The ocean is not far away.
It is downstream.**

海不遠。

海就在下游。

Thank you, Nanzhen Elementary!
謝謝南鎮國小！